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Methods and compositions for delivery of agents across the blood-brain barrier

Abstract

The present invention is based on the development of artificial targeting sequences that enhance permeation of agents into cells and across the blood brain barrier, compositions comprising the sequences, and methods of use thereof. Provided herein is an AAV comprising a capsid protein comprising a targeting sequence and a transgene, preferably a therapeutic or diagnostic transgene. Further, provided herein are methods of delivering a transgene to a cell, the method comprising contacting the cell with an AAV or fusion protein described herein.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/US2019/041386, having an International Filing Date of Jul. 11, 2019, which claims the benefit of U.S. Provisional Application Ser. No. 62/696,422, filed on Jul. 11, 2018. The entire contents of the foregoing are incorporated herein by reference.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jul. 11, 2019, is named 29618-0200W01_SL.txt and is 58,834 bytes in size.

TECHNICAL FIELD

(2) Described herein are sequences that enhance permeation of agents across the blood brain barrier, compositions comprising the sequences, and methods of use thereof.

BACKGROUND

(3) Delivery of therapeutic agents, including gene therapy reagents, is an impediment to development of treatments for a number of conditions. The blood-brain barrier (BBB) is a key obstacle for drug delivery to the mammalian central nervous system (CNS), particularly for delivery to the human brain, to treat conditions including neurodegenerative diseases such as Parkinson's disease; Alzheimer's disease; Huntington's disease; Amyotrophic lateral sclerosis; and Multiple sclerosis.

SUMMARY

(4) The present invention is based on the development of artificial targeting sequences that enhance permeation of agents into cells and across the blood brain barrier.

(5) Thus provided herein is an AAV capsid protein, e.g., an engineered AAV capsid protein, comprising a targeting sequence that comprises at least four contiguous amino acids from the sequence TVSALFK (SEQ ID NO:8); TVSALK (SEQ ID NO:4); KLASVT (SEQ ID NO:83); or KFLASVT (SEQ ID NO:84). In some embodiments, the AAV capsid protein comprises a targeting sequence that comprises at least five contiguous amino acids from the sequence TVSALK (SEQ ID NO:4); TVSALFK (SEQ ID NO:8); KLASVT (SEQ ID NO:83); or KFLASVT (SEQ ID NO: 84). In some embodiments, the AAV capsid protein comprises a targeting sequence that comprises at least six contiguous amino acids from the sequence TVSALK (SEQ ID NO:4); TVSALFK (SEQ ID NO:8); KLASVT (SEQ ID NO:83); or KFLASVT (SEQ ID NO:84).

(6) In some embodiments, the AAV is AAV9; other AAV as known in the art (e.g., AAV1, 2, 3, 4, 5, 6, 7, 8 and variants thereof and others as known in the art or described herein) can also be used.

(7) In some embodiments, the AAV capsid protein comprises AAV9 VP1 (e.g., SEQ ID NO:85).

(8) In some embodiments, the targeting sequence is inserted in the capsid protein at a position corresponding to between amino acids 588 and 589 of SEQ ID NO:85.

(9) Also provided herein are nucleic acids encoding the AAV capsid proteins comprising a targeting sequence as described herein.

(10) In addition, provided herein is an AAV comprising a capsid protein comprising a targeting sequence as described herein. In some embodiments, AAV further comprises a transgene, preferably a therapeutic or diagnostic transgene. Therapeutic transgenes can include, e.g., cDNAs that restore protein function, guide RNA for gene editing, RNA, or miRNA.

(11) Also provided herein are targeting sequences comprising V[S/p][A/m/t]/L (SEQ ID NO:79), TV[S/p][A/m/t]/L (SEQ ID NO:80), TV[S/p][A/m/t]/LK (SEQ ID NO:81), or TV[S/p][A/m/t]/LFK (SEQ ID NO:82). In some embodiments, the targeting sequence comprises VPALR (SEQ ID NO: 1); VSALK (SEQ ID NO:2); TVPALR (SEQ ID NO:3); TVSALK (SEQ ID NO:4); TVPMLK (SEQ ID NO: 12); TVPTLK (SEQ ID NO:13); FTVSALK (SEQ ID NO:5); LTVSALK (SEQ ID NO:6); TVSALFK (SEQ ID NO:8); TVPALFR (SEQ ID NO:9); TVPMLFK (SEQ ID NO:10) or TVPTLFK (SEQ ID NO:11). Also provided are fusion proteins comprising the targeting sequences linked to a heterologous (e.g., non-AAV VP1) sequence, and AAV capsid proteins (e.g., AAV9 VP1) comprising the targeting sequence. In some embodiments, the targeting sequence is inserted in a position corresponding to amino acids 588 and 589 of SEQ ID NO:85.

(12) Additionally provided herein are nucleic acids encoding the targeting sequences, fusion proteins or AAV capsid proteins described herein, as well as AAV comprising the capsid proteins comprising a targeting sequence. In some embodiments, the AAV further comprises a transgene, preferably a therapeutic or diagnostic transgene. Therapeutic transgenes can include, e.g., cDNAs that restore protein function, guide RNA for gene editing, RNA, or miRNA.

(13) Further, provided herein are methods of delivering a transgene to a cell, the method comprising contacting the cell with an AAV or fusion protein described herein. In some embodiments, the cell is in a living subject, e.g., a mammalian subject. In some embodiments, the cell is in a tissue selected from the brain, spinal cord, dorsal root ganglion, heart, or muscle, and a combination thereof. In some embodiments, the cell is a neuron (optionally a dorsal root ganglion neuron), astrocyte,

cardiomyocyte, or myocyte.

(14) In some embodiments, the subject has a neurodegenerative disease, epilepsy; stroke; spinocerebellar ataxia; Canavan's disease; Metachromatic leukodystrophy; Spinal muscular atrophy; Friedreich's ataxia; X-linked centronuclear myopathy; Lysosomal storage disease; Barth syndrome; Duchenne muscular dystrophy; Wilson's disease; or Crigler-Najjar syndrome type 1. In some embodiments, the neurodegenerative disease is Parkinson's disease; Alzheimer's disease; Huntington's disease; Amyotrophic lateral sclerosis; and Multiple sclerosis.

(15) In some embodiments, the subject has a brain cancer, and the method includes administering an AAV encoding an anti-cancer agent. In some embodiments, the anti-cancer agent is HSV.TK1, and the method further comprises administering ganciclovir.

(16) In some embodiments, the cell is in the brain of the subject, and the AAV is administered by parenteral delivery (e.g., via intravenous, intraarterial, subcutaneous, intraperitoneal, or intramuscular delivery); intracerebral; or intrathecal delivery (e.g., via lumbar injection, cisternal *magna* injection, or intraparenchymal injection).

(17) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Methods and materials are described herein for use in the present invention; other, suitable methods and materials known in the art can also be used. The materials, methods, and examples are illustrative only and not intended to be limiting. All publications, patent applications, patents, sequences, database entries, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control.

(18) Other features and advantages of the invention will be apparent from the following detailed description and figures, and from the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIGS. 1A-1B depict an exemplary strategy of engineering AAV9 by inserting cell-penetrating peptides (CPPs) into its capsid. FIG. 1A is a 3D model of an AAV9 virus. Individual CPP inserted into the capsid between amino acids 588 and 589 (VP1 numbering) will be displayed at the 3-fold axis where receptor binding presumably occurs. FIG. 1B illustrate the method of individual AAV production. Three plasmids including pRC (engineered or not), pHelper and pAAV are co-transfected into HEK 293T cells, with AAVs harvested and purified using iodixanol gradient.

(2) FIGS. 2A-2B depict representative images of mouse brain sections and their quantitative analysis after intravenous administration of low-dose candidate AAVs. Mice with mixed genetic background are used. Candidate AAVs differs in their inserted CPPs (see Table 3), but all express nuclear red fluorescent protein (RFP) as reporter. Candidate AAVs with low production yields are excluded for further screening. The dose of AAV is 1×10^{10} vg (viral genome) per animal. Each white dot in FIG. 2A represents a RFP-labeled cell. In FIG. 2B, * $P < 0.05$, vs. AAV9, ANOVA.

(3) FIGS. 2C-2D depict representative images of mouse brain sections and their quantitative analysis after intravenous administration of AAV.CPP.11 and AAV.CPP.12 in a repeat experiment. AAV.CPP.11 and AAV.CPP.12 contain CPPs BIP1 and BIP2 respectively (see Table 3). The doses of the AAVs are increased to 1×10^{11} vg per animal. Candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. Each white dot in FIG. 2C represents a RFP-labeled cell. In FIG. 2D, * $P < 0.05$, ** $P < 0.01$, vs. AAV9, ANOVA.

(4) FIG. 3A depicts the optimization of the BIP targeting sequence in order to further engineer AAV9 towards better brain transduction. BIP1 (VPALR, SEQ ID NO:1), which enables AAV9 to transduce brain more efficiently (as in AAV.CPP.11), is derived from the protein Ku70 in rats. Human, mouse and rat Ku70 proteins differ in their exact amino acid sequences. BIP2 (VSALK, SEQ ID NO:2) as in AAV.CPP.12 is a “synthetic” peptide related to BIP1. Further engineering focuses on the VSALK

sequence in the hope of minimizing species specificity of final engineered AAV. To generate new targeting sequence, amino acids of interest are added to the VSALK sequence, and in other cases positions of individual amino acids are switched. All new BIP2-derived sequences are again inserted into the AAV9 capsid to generate new candidate AAVs for screening. Sequences appearing in order are SEQ ID NOs: 69, 70, 71, 1-6, 72, 7, and 8.

(5) FIGS. 3B-3C depict representative images of mouse brain sections and their quantitative analysis after intravenous administration of more candidate AAVs. All candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. The dose of AAV is 1×10^{11} vg per animal. Each white dot in FIG. 3B represents a RFP-labeled cell. AAV.CPP.16 and AAV.CPP.21 were identified as top hits with their robust and widespread brain transduction. In FIG. 3C, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, vs. AAV9, ANOVA.

(6) FIG. 3D depicts quantitative analysis of transduction efficiency in the liver after intravenous administration of candidate AAVs. Percentage of transduced liver cells is presented. The dose of AAV is 1×10^{11} vg per animal. *** $P < 0.001$, vs. AAV9, ANOVA.

(7) FIGS. 4A-4E depict screening of selected candidate AAVs in an in vitro spheroid model of human blood-brain barrier. FIG. 4A illustrates the spheroid comprising human microvascular endothelial cells, which forms a barrier at the surface, and human pericyte and astrocytes inside the spheroid. Candidate AAVs were assessed for their ability to penetrate from the surrounding medium into the inside of the spheroid and to transduce the cells inside. FIG. 4B-4D shows images of AAV9, AAV.CPP.16 and AAV.CPP.21 treated spheroids. FIG. 4E shows relative RFP intensity of different AAV treated spheroids. *** $P < 0.001$, vs. AAV9, ANOVA.

(8) FIGS. 5A-5B depict representative images of brain sections and their quantitative analysis after intravenous administration of AAV9, AAV.CPP.16 and AAV.CPP.21 in C57BL/6J inbred mice. All candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. The dose of AAV is 1×10^{12} vg per animal. Each white dot in FIG. 5A represents a RFP-labeled cell. In FIG. 5B, * $P < 0.05$, *** $P < 0.001$, ANOVA.

(9) FIGS. 6A-6B depict representative images of brain sections and their quantitative analysis after intravenous administration of AAV9, AAV.CPP.16 and AAV.CPP.21 in BALB/cJ inbred mice. All candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. The dose of AAV is 1×10^{12} vg per animal. Each white dot in FIG. 6A represents a RFP-labeled cell. In FIG. 6B, *** $P < 0.001$, ANOVA.

(10) FIGS. 7A-7B depict representative images of brain sections and their quantitative analysis after intravenous administration of high-dose AAV.CPP.16 and AAV.CPP.21 in C57BL/6J inbred mice. Both candidate AAVs express nuclear red fluorescent protein (RFP) as reporter. The dose of AAV is 4×10^{12} vg per animal. Each white dot in FIG. 7A represents a RFP-labeled cell. In FIG. 7B, * $P < 0.05$, Student test.

(11) FIG. 8A shows AAV.CPP.16 and AAV.CPP.21 transduce adult neurons (labeled by a NeuN antibody) across multiple brain regions in mice including the cortex, midbrain and hippocampus. Transduced neurons are co-labeled by NeuN antibody and RFP. AAVs of 4×10^{12} vg were administered intravenously in adult C57BL/6J mice (6 weeks old).

(12) FIG. 8B depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 in targeting the spinal cord and motor neurons in mice. AAVs of 4×10^{10} vg were administered intravenously into neonate mice (1 day after birth). Motor neurons in the ventral horn of the spinal cord were visualized using CHAT antibody staining. Co-localization of RFP and CHAT signals suggests specific transduction of the motor neurons.

(13) FIG. 9A depicts that AAV.CPP.16 shows enhanced ability vs. AAV9 in targeting the heart in adult mice. AAVs of 1×10^{11} vg were administered intravenously in adult C57BL/6J mice (6 weeks old). Percentage of RFP-labeled cells relative to all DAPI-stained cells is presented. * $P < 0.05$, Student test.

(14) FIG. 9B depicts that AAV.CPP.16 shows enhanced ability vs. AAV9 in targeting the skeletal muscle in adult mice. AAVs of 1×10^{11} vg were administered intravenously in adult C57BL/6J

mice (6 weeks old). Percentage of RFP-labeled cells relative to all DAPI-stained cells is presented. * $P < 0.05$, Student test.

(15) FIG. 9C depicts that AAV.CPP.16 shows enhanced ability vs. AAV9 in targeting the dorsal root ganglion (DRG) in adult mice. AAVs of 1×10^{11} vg were administered intravenously in adult C57BL/6J mice (6 weeks old). Percentage of RFP-labeled cells relative to all DAPI-stained cells is presented. * $P < 0.05$, Student test.

(16) FIG. 10A depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 to transduce brain cells in primary visual cortex after intravenous administration in non-human primates. 2×10^{13} vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3 months old cynomolgus monkeys with low pre-existing neutralizing antibody. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. Squared areas in the left panels are enlarged as in the right panels. AAV.CPP.16 transduced significantly more cells vs. AAV9. AAV.CPP.21 also transduced more cell vs. AAV9 although its effect was less evident in comparison with AAV.CPP.16.

(17) FIG. 10B depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 to transduce brain cells in parietal cortex after intravenous administration in non-human primates. 2×10^{13} vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3 months old cynomolgus monkeys with low pre-existing neutralizing antibody. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. Squared areas in the left panels are enlarged as in the right panels. AAV.CPP.16 transduced significantly more cells vs. AAV9. AAV.CPP.21 also transduced more cell vs. AAV9 although its effect was less evident in comparison with AAV.CPP.16.

(18) FIG. 10C depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 to transduce brain cells in thalamus after intravenous administration in non-human primates. 2×10^{13} vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3 months old cynomolgus monkeys with low pre-existing neutralizing antibody. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. Squared areas in the left panels are enlarged as in the right panels. AAV.CPP.16 transduced significantly more cells vs. AAV9. AAV.CPP.21 also transduced more cell vs. AAV9 although its effect was less evident in comparison with AAV.CPP.16.

(19) FIG. 10D depicts that AAV.CPP.16 and AAV.CPP.21 show enhanced ability vs. AAV9 to transduce brain cells in cerebellum after intravenous administration in non-human primates. 2×10^{13} vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3 months old cynomolgus monkeys with low pre-existing neutralizing antibody. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. Squared areas in the left panels are enlarged as in the right panels. Both AAV.CPP.16 and AAV.CPP.21 transduced significantly more cells vs. AAV9.

(20) FIGS. 11A-11B depict that AAV.CPP.16 and AAV.CPP.21 do not bind to LY6A. LY6A serves as a receptor for AAV.PHP.B and its variants including AAV.PHP.eB (as in U.S. Pat. No. 9,102,949, US20170166926) and mediates AAV.PHP.eB's robust effect in crossing the BBB in certain mouse strains (Hordeaux et al. Mol Ther 2019 27(5):912-921; Huang et al. 2019, dx.doi.org/10.1101/538421). Over-expressing mouse LY6A in cultured 293 cells significantly increases binding of AAV.PHP.eB to the cell surface (FIG. 11A). On the contrary, over-expressing LY6A does not increase viral binding for AAV9, AAV.CPP.16 or AAV.CPP.21 (FIG. 11B). This suggests AAV.CPP.16 or AAV.CPP.21 does not share LY6A with AAV.PHP.eB as a receptor.

(21) FIGS. 12A-12C depict that AAV.CPP.21 can be used to systemically deliver a therapeutic gene into brain tumor in a mouse mode of glioblastoma (GBM). As in FIG. 11A, intravenously administered AAV.CPP.21-H2BmCherry was shown to target tumor mass, especially the tumor expanding frontier. In FIG. 11B-11C, using AAV.CPP.21 to systemically deliver the "suicide gene" HSV.TK1 results in shrinkage of brain tumor mass, when combined with the pro-drug ganciclovir. HSV.TK1 turns the otherwise "dormant" ganciclovir into a tumor-killing drug. * $P < 0.05$, Student test.

(22) FIG. 13 depicts that when injected locally into adult mouse brain, AAV.CPP.21 resulted in more widespread and robust transduction of brain tissue in comparison with AAV9. Intracerebral injection of AAVs (1×10^8 vg) was performed in adult mice (>6 weeks old) and brain tissues were harvested and examined 3 weeks after AAV injection. ** $P < 0.01$, Student test.

DETAILED DESCRIPTION

(23) Difficulties associated with delivery across the BBB have hindered development of therapeutic agents to treat brain disorders including cancer and neurodegenerative disorders. Adeno-associated virus (AAV) has emerged as an important research and clinical tool for delivering therapeutic genes to the brain, spinal cord and the eye; see, e.g., U.S. Pat. Nos. 9,102,949; 9,585,971; and US20170166926. However, existing AAVs including AAV9 have either limited efficiency in crossing the BBB, or only work in some non-primate species.

(24) Through rational design and targeted screening on the basis of known cell-penetrating peptides (CPPs) (see, e.g., Gomez et al., *Bax-inhibiting peptides derived from Ku70 and cell-penetrating pentapeptides*. Biochem. Soc. Trans. 2007; 35(Pt 4):797-801), targeting sequences have been discovered that, when engineered into the capsid of an AAV, improved the efficiency of gene delivery to the brain by up to three orders of magnitude. These methods were used to engineer one such AAV vector that dramatically reduced tumor size in an animal model of glioblastoma.

(25) Targeting Sequences

(26) The present methods identified a number of potential targeting peptides that enhance permeation through the BBB, e.g., when inserted into the capsid of an AAV, e.g., AAV1, AAV2, AAV8, or AAV9, or when conjugated to a biological agent, e.g., an antibody or other large biomolecule, either chemically or via expression as a fusion protein.

(27) In some embodiments, the targeting peptides comprise sequences of at least 5 amino acids. In some embodiments, the amino acid sequence comprises at least 4, e.g., 5, contiguous amino acids of the sequences VPALR (SEQ ID NO:1) and VSALK (SEQ ID NO:2).

(28) In some embodiments, the targeting peptides comprise a sequence of X.sub.1 X.sub.2 X.sub.3 X.sub.4 X.sub.5, wherein: (i) X.sub.1, X.sub.2, X.sub.3, X.sub.4 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M; and (ii) X.sub.5 is K, R, H, D, or E (SEQ ID NO:73).

(29) In some embodiments, the targeting peptides comprise sequences of at least 6 amino acids. In some embodiments, the amino acid sequence comprises at least 4, e.g., 5 or 6 contiguous amino acids of the sequences TVPALR (SEQ ID NO:3), TVSALK (SEQ ID NO:4), TVPMLK (SEQ ID NO:12) and TVPTLK (SEQ ID NO:13).

(30) In some embodiments, the targeting peptides comprise a sequence of X.sub.1 X.sub.2 X.sub.3 X.sub.4 X.sub.5 X.sub.6, wherein: (i) X.sub.1 is T; (ii) X.sub.2, X.sub.3, X.sub.4, X.sub.5 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M; and (iii) X.sub.6 is K, R, H, D, or E (SEQ ID NO:74).

(31) In some embodiments, the targeting peptides comprise a sequence of X.sub.1 X.sub.2 X.sub.3 X.sub.4 X.sub.5 X.sub.6, wherein: (i) X.sub.1, X.sub.2, X.sub.3, X.sub.4 are any four non-identical amino acids from V, A, L, I, G, P, S, T, or M; (ii) X.sub.5 is K, R, H, D, or E; and (iii) X.sub.6 is E or D (SEQ ID NO:75).

(32) In some embodiments, the targeting peptides comprise sequences of at least 7 amino acids. In some embodiments, the amino acid sequence comprises at least 4, e.g., 5, 6, or 7 contiguous amino acids of the sequences FTVSALK (SEQ ID NO:5), LTVSALK (SEQ ID NO:6), TVSALFK (SEQ ID NO:8), TVPALFR (SEQ ID NO:9), TVPMLFK (SEQ ID NO:10) and TVPTLFK (SEQ ID NO:11). In some other embodiments, the targeting peptides comprise a sequence of X.sub.1 X.sub.2 X.sub.3 X.sub.4 X.sub.5 X.sub.6 X.sub.7, wherein: (i) X.sub.1 is F, L, W, or Y; (ii) X.sub.2 is T; (iii) X.sub.3, X.sub.4, X.sub.5, X.sub.6 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M; and (iv) X.sub.7 is K, R, H, D, or E (SEQ ID NO:76).

(33) In some embodiments, the targeting peptides comprise a sequence of X.sub.1 X.sub.2 X.sub.3 X.sub.4 X.sub.5 X.sub.6 X.sub.7, wherein: (i) X.sub.1 is T; (ii) X.sub.2, X.sub.3, X.sub.4, X.sub.5 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M; (iii) X.sub.6 is K, R, H, D, or E;

and (iv) X.sub.7 is E or D (SEQ ID NO:77).

(34) In some embodiments, the targeting peptides comprise a sequence of X.sub.1 X.sub.2 X.sub.3 X.sub.4 X.sub.5 X.sub.6 X.sub.7, wherein: (i) X.sub.1, X.sub.2, X.sub.3, X.sub.4 are any four non-identical amino acids of V, A, L, I, G, P, S, T, or M; (ii) X.sub.5 is K, R, H, D, or E; (iii) X.sub.6 is E or D; and (iv) X.sub.7 is A or I (SEQ ID NO:78).

(35) In some embodiments, the targeting peptides comprise a sequence of V[S/p][A/m/t/L] (SEQ ID NO:79), wherein the upper case letters are preferred at that position. In some embodiments, the targeting peptides comprise a sequence of TV[S/p][A/m/t/L] (SEQ ID NO:80). In some embodiments, the targeting peptides comprise a sequence of TV[S/p][A/m/t/LK] (SEQ ID NO:81). In some embodiments, the targeting peptides comprise a sequence of TV[S/p][A/m/t/LFK] (SEQ ID NO:82).

(36) In some embodiments, the targeting peptide does not consist of VPALR (SEQ ID NO:1) or VSALK (SEQ ID NO:2).

(37) Specific exemplary amino acid sequences that include the above mentioned 5, 6, or 7-amino acid sequences are listed in Table 1.

(38) TABLE-US-00001 TABLE 1 Targeting Sequences SEQ ID NO: Targeting Peptide Sequence 1. VPALR 2. VSALK 3. TVPALR 4. TVSALK 5. FTVSALK 6. LTVSALK 7. TFVSALK 8. TVSALFK 9. TVPALFR 10. TVPMLFK 11. TVPTLFK 12. TVPMLK 13. TVPTLK 14. VPMLK 15. VPTLK 16. VPMLKE 17. VPTLKD 18. VPALRD 19. VSALKE 20. VSALKD 21. TAVSLK 22. TALVSK 23. TVLSAK 24. TLVSAK 25. TMVPLK 26. TMLVPK 27. TVLPMK 28. TLVPMK 29. TTVPLK 30. TTLVPK 31. TVLPTK 32. TLVPTK 33. TAVPLR 34. TALVPR 35. TVLPAR 36. TLVPAR 37. TAVSLKE 38. TALVSKE 39. TVLSAKE 40. TLVSAKE 41. TMVPLKE 42. TMLVPKE 43. TVLPMKE 44. TLVPMKE 45. TTVPLKD 46. TTLVPKD 47. TVLPTKD 48. TLVPTKD 49. TAVPLRD 50. TALVPRD 51. TVLPARD 52. TLVPARD 53. TAVSLFK 54. TALVSFK 55. TVLSAFK 56. TLVSAFK 57. TMVPLFK 58. TMLVPFK 59. TVLPMFK 60. TLVPMFK 61. TTVPLFK 62. TTLVPFK 63. TVLPTFK 64. TLVPTFK 65. TAVPLFR 66. TALVPFR 67. TVLPAFR 68. TLVPAFR

(39) Targeting peptides including reversed sequences can also be used, e.g., KLASVT (SEQ ID NO:83) and KFLASVT (SEQ ID NO:84).

(40) Targeting peptides disclosed herein can be modified according to the methods known in the art for producing peptidomimetics. See, e.g., Qvit et al., Drug Discov Today. 2017 February; 22(2): 454-462; Farhadi and Hashemian, Drug Des Devel Ther. 2018; 12: 1239-1254; Avan et al., Chem. Soc. Rev., 2014, 43, 3575-3594; Pathak, et al., Indo American Journal of Pharmaceutical Research, 2015. 8; Kazmierski, W. M., ed., Peptidomimetics Protocols, Human Press (Totowa NJ 1998); Goodman et al., eds., Houben-Weyl Methods of Organic Chemistry: Synthesis of Peptides and Peptidomimetics, Thiele Verlag (New York 2003); and Mayo et al., J. Biol. Chem., 278:45746 (2003). In some cases, these modified peptidomimetic versions of the peptides and fragments disclosed herein exhibit enhanced stability in vivo, relative to the non-peptidomimetic peptides.

(41) Methods for creating a peptidomimetic include substituting one or more, e.g., all, of the amino acids in a peptide sequence with D-amino acid enantiomers. Such sequences are referred to herein as “retro” sequences. In another method, the N-terminal to C-terminal order of the amino acid residues is reversed, such that the order of amino acid residues from the N-terminus to the C-terminus of the original peptide becomes the order of amino acid residues from the C-terminus to the N-terminus in the modified peptidomimetic. Such sequences can be referred to as “inverso” sequences.

(42) Peptidomimetics can be both the retro and inverso versions, i.e., the “retro-inverso” version of a peptide disclosed herein. The new peptidomimetics can be composed of D-amino acids arranged so that the order of amino acid residues from the N-terminus to the C-terminus in the peptidomimetic corresponds to the order of amino acid residues from the C-terminus to the N-terminus in the original peptide.

(43) Other methods for making a peptidomimetic include replacing one or more amino acid residues in a peptide with a chemically distinct but recognized functional analog of the amino acid, i.e., an

artificial amino acid analog. Artificial amino acid analogs include β -amino acids, β -substituted β -amino acids (“ β .sup.3-amino acids”), phosphorous analogs of amino acids, such as ∇ -amino phosphonic acids and ∇ -amino phosphinic acids, and amino acids having non-peptide linkages. Artificial amino acids can be used to create peptidomimetics, such as peptoid oligomers (e.g., peptoid amide or ester analogues), β -peptides, cyclic peptides, oligourea or oligocarbamate peptides; or heterocyclic ring molecules. Exemplary retro-inverso targeting peptidomimetics include KLASVT and KFLASVT, wherein the sequences include all D-amino acids. These sequences can be modified, e.g., by biotinylation of the amino terminus and amidation of the carboxy terminus.

(44) AAVs

(45) Viral vectors for use in the present methods and compositions include recombinant retroviruses, adenovirus, adeno-associated virus, alphavirus, and lentivirus, comprising the targeting peptides described herein and optionally a transgene for expression in a target tissue.

(46) A preferred viral vector system useful for delivery of nucleic acids in the present methods is the adeno-associated virus (AAV). AAV is a tiny non-enveloped virus having a 25 nm capsid. No disease is known or has been shown to be associated with the wild type virus. AAV has a single-stranded DNA (ssDNA) genome. AAV has been shown to exhibit long-term episomal transgene expression, and AAV has demonstrated excellent transgene expression in the brain, particularly in neurons. Vectors containing as little as 300 base pairs of AAV can be packaged and can integrate. Space for exogenous DNA is limited to about 4.7 kb. An AAV vector such as that described in Tratschin et al., Mol. Cell. Biol. 5:3251-3260 (1985) can be used to introduce DNA into cells. A variety of nucleic acids have been introduced into different cell types using AAV vectors (see for example Hermonat et al., Proc. Natl. Acad. Sci. USA 81:6466-6470 (1984); Tratschin et al., Mol. Cell. Biol. 4:2072-2081 (1985); Wondisford et al., Mol. Endocrinol. 2:32-39 (1988); Tratschin et al., J. Virol. 51:611-619 (1984); and Flotte et al., J. Biol. Chem. 268:3781-3790 (1993). There are numerous alternative AAV variants (over 100 have been cloned), and AAV variants have been identified based on desirable characteristics. In some embodiments, the AAV is AAV1, AAV2, AAV4, AAV5, AAV6, AV6.2, AAV7, AAV8, AAV9, rh.10, rh.39, rh.43 or CSp3; for CNS use, in some embodiments the AAV is AAV1, AAV2, AAV4, AAV5, AAV6, AAV8, or AAV9. As one example, AAV9 has been shown to somewhat efficiently cross the blood-brain barrier. Using the present methods, the AAV capsid can be genetically engineered to increase permeation across the BBB, or into a specific tissue, by insertion of a targeting sequence as described herein into the capsid protein, e.g., into the AAV9 capsid protein VP1 between amino acids 588 and 589.

(47) An exemplary wild type AAV9 capsid protein VP1 (Q6JC40-1) sequence is as follows:

(48) TABLE-US-00002 (SEQ ID NO: 85)

				10	
20	30	40	50	MAADGYLPDW	
LEDNLSEGIR	EWALKPGAP	QPKANQQHQD	NARGLVLPGY		
60	70	80	90	100	
KYLGPNGLD	KGEPVNAADA	AALEHDKAYD	QLLKAGDNPY	LKYNHADAEF	
	110	120	130		
140	150	QERLKEDTSF	GGNLGRAVFQ	AKKRILLEPLG	
LVEEAAKTAP	GKKRPVEQSP		160	170	
180	190	200	QEPDSSAGIG	KSGAQPAKKR	
LNFGQTGDTE	SVPDPQPIGE	PPAAPSGVGS	210		
220	230	240	250	LTMASGGGAP	
VADNNEGADG	VGSSSGNWHC	DSQWLGDRVI	TTSTRTWALP		
260	270	280	290	300	
TYNNHLYKQI	SNSTSGGSSN	DNAYFGYSTP	WGYFDFNRFH	CHESPRDWQR	
	310	320	330		
340	350	LINNNWGRFP	KRLNFKLFNI	QVKEVTDNNG	
VKTIANNLTS	TVQVFTDSY		360	370	
380	390	400	QLPYVLGSAH	EGCLPPFPAD	

VEMIPQYGYL	TLNDGSQAVG	RSSFYCLEYF	410
420	430	440	450 PSQMLRTGNN
FQFSYEFENV	PFHSSYAHSQ	SLDRLMNPLI	DQYLYYLSKT
460	470	480	490 500
INGSGQNQQT	LKFSVAGPSN	MAVQGRNYIP	GPSYRQQRVS TTVTQNNNSE
510	520	530	
540	550 FAWPGASSWA	LNGRNSLMNP	GPAMASHKEG
EDRFFPLSGS	LIFGKQGTGR	560	570
580	590	600 DNVDADKVM	TNEEEIKTTN
PVATESYGQV	ATNHQSAQAQ	AQTGWVQNQG	610
620	630	640	650 ILPGMVWQDR
DVYLQGPIWA	KIPHTDGNFH	PSPLMGGFGM	KHPPPQILIK
660	670	680	690 700
NTPVPADPPT	AFNKDKLNSF	ITQYSTGQVS	VEIEWELQKE NSKRWNPEIQ
710	720	730 YTSNYYKSNN	
VEFAVNTEGV	YSEPRPIGTR	YLTRNL	

(49) Thus provided herein are AAV that include one or more of the targeting peptide sequences described herein, e.g., an AAV comprising a capsid protein comprising a targeting sequence described herein, e.g., a capsid protein comprising SEQ ID NO:1 wherein a targeting peptide sequence has been inserted into the sequence, e.g., between amino acids 588 and 589.

(50) In some embodiments, the AAV also includes a transgene sequence (i.e., a heterologous sequence), e.g., a transgene encoding a therapeutic agent, e.g., as described herein or as known in the art, or a reporter protein, e.g., a fluorescent protein, an enzyme that catalyzes a reaction yielding a detectable product, or a cell surface antigen. The transgene is preferably linked to sequences that promote/drive expression of the transgene in the target tissue.

(51) Exemplary transgenes for use as therapeutics include neuronal apoptosis inhibitory protein (NAIP), nerve growth factor (NGF), glial-derived growth factor (GDNF), brain-derived growth factor (BDNF), ciliary neurotrophic factor (CNTF), tyrosine hydroxylase (TH), GTP-cyclohydrolase (GTPCH), amino acid decarboxylase (AADC), aspartoacylase (ASPA), blood factors, such as β -globin, hemoglobin, tissue plasminogen activator, and coagulation factors; colony stimulating factors (CSF); interleukins, such as IL-1, IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, etc.; growth factors, such as keratinocyte growth factor (KGF), stem cell factor (SCF), fibroblast growth factor (FGF, such as basic FGF and acidic FGF), hepatocyte growth factor (HGF), insulin-like growth factors (IGFs), bone morphogenetic protein (BMP), epidermal growth factor (EGF), growth differentiation factor-9 (GDF-9), hepatoma derived growth factor (HDGF), myostatin (GDF-8), nerve growth factor (NGF), neurotrophins, platelet-derived growth factor (PDGF), thrombopoietin (TPO), transforming growth factor alpha (TGF- α), transforming growth factor beta (TGF- β), and the like; soluble receptors, such as soluble TNF- α receptors, soluble VEGF receptors, soluble interleukin receptors (e.g., soluble IL-1 receptors and soluble type II IL-1 receptors), soluble gamma/delta T cell receptors, ligand-binding fragments of a soluble receptor, and the like; enzymes, such as α -glucosidase, imiglucrase, β -glucocerebrosidase, and alglucerase; enzyme activators, such as tissue plasminogen activator; chemokines, such as IP-10, monokine induced by interferon-gamma (Mig), Groa/IL-8, RANTES, MIP-1 α , MIP-10, MCP-1, PF-4, and the like; angiogenic agents, such as vascular endothelial growth factors (VEGFs, e.g., VEGF121, VEGF165, VEGF-C, VEGF-2), transforming growth factor-beta, basic fibroblast growth factor, glioma-derived growth factor, angiogenin, angiogenin-2; and the like; anti-angiogenic agents, such as a soluble VEGF receptor; protein vaccine; neuroactive peptides, such as nerve growth factor (NGF), bradykinin, cholecystokinin, gastrin, secretin, oxytocin, gonadotropin-releasing hormone, beta-endorphin, enkephalin, substance P, somatostatin, prolactin, galanin, growth hormone-releasing hormone, bombesin, dynorphin, warfarin, neurotensin, motilin, thyrotropin, neuropeptide Y, luteinizing hormone, calcitonin, insulin, glucagons, vasopressin, angiotensin II, thyrotropin-releasing hormone, vasoactive intestinal peptide, a sleep peptide, and the like;

thrombolytic agents; atrial natriuretic peptide; relaxin; glial fibrillary acidic protein; follicle stimulating hormone (FSH); human alpha-1 antitrypsin; leukemia inhibitory factor (LIF); transforming growth factors (TGFs); tissue factors, luteinizing hormone; macrophage activating factors; tumor necrosis factor (TNF); neutrophil chemotactic factor (NCF); nerve growth factor; tissue inhibitors of metalloproteinases; vasoactive intestinal peptide; angiogenin; angiotropin; fibrin; hirudin; IL-1 receptor antagonists; and the like. Some other examples of protein of interest include ciliary neurotrophic factor (CNTF); neurotrophins 3 and 4/5 (NT-3 and 4/5); glial cell derived neurotrophic factor (GDNF); aromatic amino acid decarboxylase (AADC); hemophilia related clotting proteins, such as Factor VIII, Factor IX, Factor X; dystrophin or mini-dystrophin; lysosomal acid lipase; phenylalanine hydroxylase (PAH); glycogen storage disease-related enzymes, such as glucose-6-phosphatase, acid maltase, glycogen debranching enzyme, muscle glycogen phosphorylase, liver glycogen phosphorylase, muscle phosphofructokinase, phosphorylase kinase (e.g., PHKA2), glucose transporter (e.g., GLUT2), aldolase A, β -enolase, and glycogen synthase; lysosomal enzymes (e.g., beta-N-acetylhexosaminidase A); and any variants thereof.

(52) The transgene can also encode an antibody, e.g., an immune checkpoint inhibitory antibody, e.g., to PD-L1, PD-1, CTLA-4 (Cytotoxic T-Lymphocyte-Associated Protein-4; CD152); LAG-3 (Lymphocyte Activation Gene 3; CD223); TIM-3 (T-cell Immunoglobulin domain and Mucin domain 3; HAVCR2); TIGIT (T-cell Immunoreceptor with Ig and ITIM domains); B7-H3 (CD276); VSIR (V-set immunoregulatory receptor, aka VISTA, B7H5, C10orf54); BTLA 30 (B- and T-Lymphocyte Attenuator, CD272); GARP (Glycoprotein A Repetitions; Predominant; PVRIG (PVR related immunoglobulin domain containing); or VTCN1 (Vset domain containing T cell activation inhibitor 1, aka B7-H4).

(53) Other transgenes can include small or inhibitory nucleic acids that alter/reduce expression of a target gene, e.g., siRNA, shRNA, miRNA, antisense oligos, or long non-coding RNAs that alter gene expression (see, e.g., WO2012087983 and US20140142160), or CRISPR Cas9/cas12a and guide RNAs.

(54) The virus can also include one or more sequences that promote expression of a transgene, e.g., one or more promoter sequences; enhancer sequences, e.g., 5' untranslated region (UTR) or a 3' UTR; a polyadenylation site; and/or insulator sequences. In some embodiments, the promoter is a brain tissue specific promoter, e.g., a neuron-specific or glia-specific promoter. In certain embodiments, the promoter is a promoter of a gene selected to from: neuronal nuclei (NeuN), glial fibrillary acidic protein (GFAP), MeCP2, adenomatous polyposis *coli* (APC), ionized calcium-binding adapter molecule 1 (Iba-1), synapsin I (SYN), calcium/calmodulin-dependent protein kinase II, tubulin alpha I, neuron-specific enolase and platelet-derived growth factor beta chain. In some embodiments, the promoter is a pan-cell type promoter, e.g., cytomegalovirus (CMV), beta glucuronidase, (GUSB), ubiquitin C (UBC), or rous sarcoma virus (RSV) promoter. The woodchuck hepatitis virus posttranscriptional response element (WPRE) can also be used.

(55) In some embodiments, the AAV also has one or more additional mutations that increase delivery to the target tissue, e.g., the CNS, or that reduce off-tissue targeting, e.g., mutations that decrease liver delivery when CNS, heart, or muscle delivery is intended (e.g., as described in Pulicherla et al. (2011) Mol Ther 19:1070-1078); or the addition of other targeting peptides, e.g., as described in Chen et al. (2008) Nat Med 15:1215-1218 or Xu et al., (2005) Virology 341:203-214 or U.S. Pat. Nos. 9,102,949; 9,585,971; and US20170166926. See also Gray and Samulski (2011) "Vector design and considerations for CNS applications," in Gene Vector Design and Application to Treat Nervous System Disorders ed. Glorioso J., editor. (Washington, DC: Society for Neuroscience;) 1-9, available at [sfn.org/~media/SfN/Documents/Short %20Courses/20110%20Short %20Course %20I/2011_SC1_Gray.ashx](http://sfn.org/~media/SfN/Documents/Short%20Courses/20110%20Short%20Course%20I/2011_SC1_Gray.ashx).

(56) Targeting Peptides as Tags/Fusions

(57) The targeting peptides described herein can also be used to increase permeation of other (heterologous) molecules across the BBB, e.g., by conjugation to the molecule, or by expression as part of a fusion protein, e.g., with an antibody or other large biomolecule. These can include genome

editing proteins or complexes (e.g., TALEs, ZFNs, Base editors, and CRISPR RNPs comprising a gene editing protein such as Cas9 or Cas12a, fused to a peptide described herein (e.g., at the N terminus, C terminus, or internally) and a guide RNA), in addition to therapeutic agents or reporters as described herein as well as those listed in Table 2. The fusions/complexes do not comprise any other sequences from Ku70, e.g., comprise heterologous non-Ku70 sequences, and are not present in nature.

(58) In some embodiments, targeting sequences used as part of a non-AAV fusion protein do not comprise or consist of VPALR (SEQ ID NO:1) or VSALK (SEQ ID NO:2).

(59) Methods of Use

(60) The methods and compositions described herein can be used to deliver any composition, e.g., a sequence of interest to a tissue, e.g., to the central nervous system (brain), heart, muscle, or dorsal root ganglion or spinal cord (peripheral nervous system). In some embodiments, the methods include delivery to specific brain regions, e.g., cortex, cerebellum, hippocampus, substantia nigra, amygdala. In some embodiments, the methods include delivery to neurons, astrocytes, glial cells, or cardiomyocytes.

(61) In some embodiments, the methods and compositions, e.g., AAVs, are used to deliver a nucleic acid sequence to a subject who has a disease, e.g., a disease of the CNS; see, e.g., U.S. Pat. Nos. 9,102,949; 9,585,971; and US20170166926. In some embodiments, the subject has a condition listed in Table 2; in some embodiments, the vectors are used to deliver a therapeutic agent listed in Table 2 for treating the corresponding disease listed in Table 2. The therapeutic agent can be delivered as a nucleic acid, e.g. via a viral vector, wherein the nucleic acid encodes a therapeutic protein or other nucleic acid such as an antisense oligo, siRNA, shRNA, and so on; or as a fusion protein/complex with a targeting peptide as described herein.

(62) TABLE-US-00003 TABLE 2 Diseases Examples of diseases Tissue targeted Therapeutic agent
Parkinson's disease CNS GDNF, AADC Alzheimer's disease CNS Tau antibody, APP antibody
Huntington's disease CNS miRNA targeting HTT Amyotrophic lateral sclerosis CNS shRNA targeting
SOD Multiple sclerosis CNS IFN-beta Epilepsy CNS Neuropeptide Y Stroke CNS IGF-1,
osteopontin Brain cancer CNS HSV.TK1, PD-1/PD-L1 antibody Spinocerebellar ataxia CNS RNAi
targeting ataxin Canavan disease CNS ASPA Metachromatic Nervous systems ARSA, PSAP
leukodystrophy Spinal muscular Neuromuscular system SMN1 atrophy Friedreich's ataxia Nervous
systems, heart Frataxin X-linked myotubular Neuromuscular system MTM1 myopathy Pompe
disease Lysosome (global GAA including CNS) Barth syndrome Heart, muscle TAZ Duchenne
muscular Muscle dystrophin dystrophy Wilson's disease Brain, liver ATP7B Crigler-Najjar Liver
UGT1A1 syndrome type 1

(63) In some embodiments, the compositions and methods are used to treat brain cancer. Brain cancers include gliomas (e.g., glioblastoma multiforme (GBM)), metastases (e.g., from lung, breast, melanoma, or colon cancer), meningiomas, pituitary adenomas, and acoustic neuromas. The compositions include a targeting peptide linked to an anticancer agent, e.g., a “suicide gene” that induces apoptosis in a target cell (e.g., HSV.TK1, Cytosine Deaminase (CD) from Herpes simplex virus or *Escherichia coli*, or *Escherichia coli* purine nucleoside phosphorylase (PNP)/fludarabine; see Krohne et al., Hepatology. 2001 September; 34(3):511-8; Dey and Evans, “Suicide Gene Therapy by Herpes Simplex Virus-1 Thymidine Kinase (HSV-TK)” (2011) DOI: 10.5772/18544) an immune checkpoint inhibitory antibody as known in the art or described herein. For example, an AAV vector comprising a targeting peptide as described herein can be used to deliver the “suicide gene” HSV.TK1 to a brain tumor. HSV.TK1 turns the otherwise “dormant” ganciclovir into a tumor-killing drug. Thus the methods can include systemically, e.g., intravenously, administering an AAV (e.g., AAV9) comprising a targeting peptide as described herein and encoding HSV.TK1, and the pro-drug ganciclovir, to a subject who has been diagnosed with brain cancer.

(64) Pharmaceutical Compositions and Methods of Administration

(65) The methods described herein include the use of pharmaceutical compositions comprising the targeting peptides as an active ingredient.

(66) Pharmaceutical compositions typically include a pharmaceutically acceptable carrier. As used herein the language “pharmaceutically acceptable carrier” includes saline, solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration.

(67) Pharmaceutical compositions are typically formulated to be compatible with its intended route of administration. Examples of routes of administration include parenteral, e.g., intravenous, intraarterial, subcutaneous, intraperitoneal intramuscular or injection or infusion administration. Delivery can thus be systemic or localized.

(68) Methods of formulating suitable pharmaceutical compositions are known in the art, see, e.g., *Remington: The Science and Practice of Pharmacy*, 21st ed., 2005; and the books in the series *Drugs and the Pharmaceutical Sciences: a Series of Textbooks and Monographs* (Dekker, NY). For example, solutions or suspensions used for parenteral application can include the following components: a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerine, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as ethylenediaminetetraacetic acid; buffers such as acetates, citrates or phosphates and agents for the adjustment of tonicity such as sodium chloride or dextrose. pH can be adjusted with acids or bases, such as hydrochloric acid or sodium hydroxide. The parenteral preparation can be enclosed in ampoules, disposable syringes or multiple dose vials made of glass or plastic.

(69) Pharmaceutical compositions suitable for injectable use can include sterile aqueous solutions (where water soluble) or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersion. For intravenous administration, suitable carriers include physiological saline, bacteriostatic water, Cremophor EL™ (BASF, Parsippany, NJ) or phosphate buffered saline (PBS). In all cases, the composition must be sterile and should be fluid to the extent that easy syringability exists. It should be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), and suitable mixtures thereof. The proper fluidity can be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. Prevention of the action of microorganisms can be achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol, sodium chloride in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent that delays absorption, for example, aluminum monostearate and gelatin.

(70) Sterile injectable solutions can be prepared by incorporating the active compound in the required amount in an appropriate solvent with one or a combination of ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the active compound into a sterile vehicle, which contains a basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum drying and freeze-drying, which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.

(71) In one embodiment, the therapeutic compounds are prepared with carriers that will protect the therapeutic compounds against rapid elimination from the body, such as a controlled release formulation, including implants and microencapsulated delivery systems. Biodegradable, biocompatible polymers can be used, such as ethylene vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, and polylactic acid. Such formulations can be prepared using standard techniques, or obtained commercially, e.g., from Alza Corporation and Nova Pharmaceuticals, Inc. Liposomal suspensions (including liposomes targeted to selected cells with monoclonal antibodies to

cellular antigens) can also be used as pharmaceutically acceptable carriers. These can be prepared according to methods known to those skilled in the art, for example, as described in U.S. Pat. No. 4,522,811.

(72) The pharmaceutical compositions can be included in a kit, container, pack, or dispenser together with instructions for administration. For example, the compositions comprising an AAV comprising a targeting peptide as described herein and a nucleic acid encoding HSV.TK1 can be provided in a kit with ganciclovir.

EXAMPLES

(73) The invention is further described in the following examples, which do not limit the scope of the invention described in the claims.

Materials and Methods

(74) The following materials and methods were used in the Examples below.

(75) 1. Generation of Capsid Variants

(76) To generate the capsid variant plasmids, DNA fragments that encode the cell-penetrating peptides (Table 3) were synthesized (GenScript), and inserted into the backbone of the AAV9 Rep-cap plasmid (pRC9) between amino acid position 588 and 589 (VP1 amino acid numbering), using CloneEZ seamless cloning technology (GenScript). CPPs BIP1(VPALR, SEQ ID NO:1) and BIP2 (VSALK, SEQ ID NO:2), as well as their derivatives such as TVSALK (SEQ ID NO:4) in AAV.CPP.16 and TVSALFK (SEQ ID NO:8) in AAV.CPP.21, are derived from the Ku70 proteins, of which the sequences are provided as below:

(77) TABLE-US-00004 Human Ku70

MSGWESYYKTEGDEEAEEEEQEENLEASGDYKYSGRDSLIFLVDASKAMFESQSEDELTPF

60 Mouse Ku70 MSEWESYYKTEGEEEEEE--

EESPD TGGEYKYSGRDSLIFLVDASRAMFESQGEDELTPF 58 Rat Ku70

MSEWESYYKTEGEEEEEE--EQSPDTNGEYKYSGRDSLIFLVDASRAMFESQGEDELTPF

58 Human Ku70

DMSIQCIQSVYISKIISDRDLLAVVFGTEKDKNSVNFKNYVLQELDNPGAKRILELD 120

Mouse Ku70

DMSIQCIQSVYTSKIISDRDLLAVVFGTEKDKNSVNFKNYVLQDLNPGAKRVLELD

118 Rat Ku70

DMSIQCIQSVYTSKIISDRDLLAVVFGTEKDKNSVNFKSIYVLQDLNPGAKRVLELD

118 Human Ku70

QFKGQQGQKRFQDMMGHGSDYSLSEVLWVCANLFSDVQFKMSHKRIMLFTNEDNPHGNDS

180 Mouse Ku70

QFKGQQGKKHFRDTVGHGSDYSLSEVLWVCANLFSDVQLKMSHKRIMLFTNEDDPHGRDS

178 Rat Ku70

RFKGQQGKKHFRDTIGHGSDYSLSEVLWVCANLFSDVQFKMSHKRIMLFTNEDDPHGNDS

178 Human Ku70

AKASRARTKAGDLRDTGIFLDMHLKKPGGFDISLFYRDIISIAEDEDLRVHFEESKLE 240

Mouse Ku70

AKASRARTKASDLRDTGIFLDMHLKKPGGFDVSVFYRDIITTAEDEDLGVHFEESKLE

238 Rat Ku70

AKASRARTKASDLRDTGIFLDMHLKKRGGFDVSLFYRDIISIAEDEDLGVHFEESKLE

238 Human Ku70

DLLRKVRACKETRKALSRLKLKLNKDVISVGIYNLVQKALKPPPIKLYRETNEPVKTKT

300 Mouse Ku70

DLLRKVRACKETKKRVLSRLKFKLGEDVVL MVGIYNLVQKANKPFPVRLYRETNEPVKTKT

298 Rat Ku70

DLLRKVRACKETKKRVLSRLKFKLGKDVAL MVGVYNLVQKANKPFPVRLYRETNEPVKTKT

298 Human Ku70

RTFNTSTGGLLLPSDTKRSQIYGSRQIILEKEETEELKRFDDPGLMLMGFKPLVLLKKHH 360

Mouse Ku70
 RTFNVNTGSLLLPSDTKRSLTYGTRQIVLEKEETEELKRFDEPGLILMGFKPTVMLKKQH
 358 Rat Ku70
 RTFNVNTGSLLLPSDTKRSLTFGTRQIVLEKEETEELKRFDEPGLILMGFKPMVMLKNHH
 358 Human Ku70
 YLRPSLFVYPEESLVIGSSTLFSALLIKCLEKEVAALCRYTPRRNIPPYFVALVPQEEEL 420
 Mouse Ku70
 YLRPSLEVYPEESLVSGSSTLFSALLTKCVEKEVIAVCRYTPRKNVSPYFVALVPQEEEL 418
 Rat Ku70
 YLRPSLFLYPEESLVNGSSTLFSALLTKCVEKEVIAVCRYTARKNVSPYFVALVPQEEEL 418
 Human Ku70
 DDQKIQVTPPGFQLVFLPFADDDKRKMPFTEKIMATPEQVGKMKAIVEKLRFTYRSDSFEN
 480 Mouse Ku70
 DDQNIQVTPGGFQLVFLPYADDDKRKVPFTEKVTANQEQIDKMKAIVQKLRFTYRSDSFEN
 478 Rat Ku70
 DDQNIQVTPAGFQLVFLPYADDDKRKVPFTEKVMANPEQIDKMKAIVQKLRFTYRSDSFEN
 478 Human Ku70
 PVLQQHFRNLEALALDLMEPEQAVDLTLPKVEAMNKRLGSLVDEFKELVYPPDYNPEGKV
 540 Mouse Ku70
 PVLQQHFRNLEALALDMMESEQVVDLTLPKVEAIKKRLGSLADEFKELVYPPGYNPEGKV
 538 Rat Ku70
 PVLQQHFRNLEALALDMMESEQVVDLTLPKVEAIKKRLGSLADEFKELVYPPGYNPEGKI
 538 Human Ku70
 TKRKHDNEGSGSKRPKVEYSEEELKTHISKGTLGKFTVPMLKEACRAYGLKSGLKQELL
 600 Mouse Ku70
 AKRKQDDEGSTSKPKVELSEEELKAHFRKGTLGKLTVP TLKDICKAHGLKSGPKKQELL
 598 Rat Ku70
 AKRKADNEGSAASKPKVELSEEELKDLFAKGTLGKLTVPALRDICKAYGLKSGPKKQELL
 598 Human Ku70
 EALT KHFQD- 609 (SEQ ID NO: 86) Mouse Ku70
 DALIRHLEKN 608 (SEQ ID NO: 87) Rat Ku70
 EALS RHLEKN 608 (SEQ ID NO: 88)
 (78) In addition, VP1 protein sequences for AAV9, AAV.CPP.16 and AAV.CPP.21 are provided as below:
 (79) TABLE-US-00005 AAV9
 MAADGYLPDWLEDNLSEGIREWWALKPGAPQPKANQQHQDNARGLVLPGYKYLGPNGLD
 60 AAV.CPP16
 MAADGYLPDWLEDNLSEGIREWWALKPGAPQPKANQQHQDNARGLVLPGYKYLGPNGLD
 60 AAV.CPP21
 MAADGYLPDWLEDNLSEGIREWWALKPGAPQPKANQQHQDNARGLVLPGYKYLGPNGLD
 60 AAV9
 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADA EFQERLKEDTSFGGNLGRAVEQ
 120 AAV.CPP16
 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADA EFQERLKEDTSFGGNLGRAVFQ
 120 AAV.CPP21
 KGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADA EFQERLKEDTSFGGNLGRAVEQ
 120 AAV9
 AKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIGKSGAQPAKKRLNFGQTGDTE
 180 AAV.CPP16
 AKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIGKSGAQPAKKRLNFGQTGDTE
 180 AAV.CPP21
 AKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIGKSGAQPAKKRLNFGQTGDTE

180 AAV9
SVPDPQPIGEPPAAPSGVGSLTMASGGGAPVADNNEGADGVGSSSGNWHCDSQWLGDRI
240 AAV.CPP16
SVPDPQPIGEPPAAPSGVGSLTMASGGGAPVADNNEGADGVGSSSGNWHCDSQWLGDRI
240 AAV.CPP21
SVPDPQPIGEPPAAPSGVGSLTMASGGGAPVADNNEGADGVGSSSGNWHCDSQWLGDRI
240 AAV9
TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFHCHFSPRDWQR
300 AAV.CPP16
TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFHCHFSPRDWQR
300 AAV.CPP21
TTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFHCHFSPRDWQR
300 AAV9
LINNNWGFPRKRLNFKLFNIQVKEVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVLGSAH
360 AAV.CPP16
LINNNWGFPRKRLNFKLFNIQVKEVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVLGSAH
360 AAV.CPP21
LINNNWGFPRKRLNFKLFNIQVKEVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVLGSAH
360 AAV9
EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYFPSQMLRTGNNFQFSYEFENV
420 AAV.CPP16
EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYFPSQMLRTGNNFQFSYEFENV
420 AAV.CPP21
EGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYFPSQMLRTGNNFQFSYEFENV
420 AAV9
PFHSSYAHSQSLDRLMNPLIDQYLYYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP
480 AAV.CPP16
PFHSSYAHSQSLDRLMNPLIDQYLYYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP
480 AAV.CPP21
PFHSSYAHSQSLDRLMNPLIDQYLYYLSKTINGSGQNQQTLKFSVAGPSNMAVQGRNYIP
480 AAV9
GPSYRQQRVSTTVTQNNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEGEDRFFPLSGS
540 AAV.CPP16
GPSYRQQRVSTTVTQNNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEGEDRFFPLSGS
540 AAV.CPP21
GPSYRQQRVSTTVTQNNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEGEDRFFPLSGS
540 AAV9 LIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVATNHQSAQ-----
AQAQT 593 AAV.CPP16
LIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVATNHQSAQTVSAL-KAQAQT
599 AAV.CPP21
LIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVATNHQSAQTVSALFKAQAQT
600 AAV9
GWVQNQGILPGMVWQDRDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMKHPPPQILIKNTP
653 AAV.CPP16
GWVQNQGILPGMVWQDRDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMKHPPPQILIKNTP
659 AAV.CPP21
GWVQNQGILPGMVWQDRDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMKHPPPQILIKNTP
660 AAV9
VPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNVEF
713 AAV.CPP16
VPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNVEF

719 AAV.CPP21

VPADPPTAFNKKDKLNSFITQYSTGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSNNVEF

720 AAV9 AVNTEGVYSEPRPIGTRYLTRNL 736 (SEQ ID NO: 85) AAV.CPP16

AVNTEGVYSEPRPIGTRYLTRNL 742 (SEQ ID NO: 89) AAV.CPP21

AVNTEGVYSEPRPIGTRYLTRNL 743 (SEQ ID NO: 90)

2. Recombinant AAV Production

(80) Recombinant AAVs were packaged using standard three-plasmid co-transfection protocol (pRC plasmid, pHelper plasmid and pAAV plasmid). pRC9 (or its variant), pHelper and pAAV carrying a transgene (e.g. nucleus-directed RFP H2B-mCherry driven by an ubiquitous EF1a promoter) were co-transfected into HEK 293T cells using polyethylenimine (PEI, Polysciences). rAAVs vectors were collected from serum-free medium 72 h and 120 h post transfection and from cell at 120 h post transfection. AAV particles in the medium were concentrated using a PEG-precipitation method with 8% PEG-8000 (wt/vol). Cell pellets containing viral particles were resuspended and lysed through sonication. Combined viral vectors from PEG-precipitation and cell lysates were treated with DNase and RNase at 37° C. for 30 mins and then purified by iodixanol gradient (15%, 25%, 40% and 60%) with ultracentrifugation (VTi 50 rotor, 40,000 r.p.m, 18° C., 1 h). rAAVs were then concentrated using Millipore Amicon filter unit (UFC910008, 100K MWCO) and formulated in Dulbecco's phosphate buffered saline (PBS) containing 0.001% Pluronic F68 (Gibco).

(81) 3. AAV Titering

(82) Virus titer was determined by measuring DNase-resistant genome copies using quantitative PCR. pAAV-CAG-GFP was digested with PVUII(NEB) to generate free ends for the plasmid ITRs, and was used for generating a standard curve. Virus samples were incubated with DNase I to eliminate contaminating DNA, followed by sodium hydroxide treatment to dissolve the viral capsid and to release the viral genome. Quantitative PCR was performed using an ITR Forward primer 5'-GGAACCCCTAGTGATGGAGTT (SEQ ID NO:91) and an ITR Reverse primer 5'-CGGCCTCAGTGAGCGA (SEQ ID NO:92). Vector titers were normalized to the rAAV-2 reference standard materials (RSMs, ATCC, cat No:VR-1616, Manassas, VA).

(83) 4. Administration of AAV in Mice

(84) For intravenous administration, AAV diluted in sterile saline (0.2 ml) was administered through tail vein injection in adult mice (over 6 weeks of age). Animals then survived for three weeks before being euthanized for tissue harvesting. For intracerebral injection, AAV diluted in PBS (10 µl) was injected using a Hamilton syringe with coordinates from bregma: 1.0 mm right, 0.3 backward, 2.6 mm deep. All animal studies were performed in an AAALAC-accredited facility with IACUC approval.

(85) 5. Mouse Tissue Processing

(86) Anesthetized animals were transcardially perfused with cold phosphate buffered saline (PBS) followed by 4% paraformaldehyde (PFA). Tissues were post-fixed in 4% PFA overnight, and then immersed in 30% sucrose solutions for two days prior to embedding and snap-freezing in OCT. Typically, 80 µm thick brain sections were cut for imaging of native fluorescence, 40 µm thick brain sections for IHC.

(87) 6. In Vitro Human BBB Spheroid Model

(88) Hot 1% agarose (w/v, 50 µl) was added in a 96-well plate to cool/solidify. Primary human astrocytes (Lonza Bioscience), human brain microvascular pericytes (HBVP, ScienCell Research Laboratories) and human cerebral microvascular endothelial cells (hCMEC/D3; Cedarlane) were then seeded onto the agarose gel in a 1:1:1 ratio (1500 cells of each type). Cells were cultured at 37° C. in a 5% CO₂ incubator for 48-72 hours to allow for spontaneous assembly of multicellular BBB spheroids. A multicellular barrier was reported to form at the periphery of the spheroid, mimicking the BBB. AAVs-H2B-mCherry were added to the culture medium, and 4 days later all spheroids were fixed using 4% PFA, transferred into a Nunc Lab-Tek II thin-glass 8-well chambered coverglass (Thermo Scientific), and imaged using a Zeiss LSM710 confocal microscope. The intensity of RFP signal inside the spheroids was examined and used as a “read-out”.

(89) 7. AAV Administration in Non-Human Primate (NHP)

(90) All NHP studies were performed by a CRO in an AAALAC-accredited facility with IACUC approval. Cynomolgus monkeys were pre-screened for little or no pre-existing neutralizing antibody against AAV9 (titer of <1:5). AAV diluted in PBS/0.001% F68 was injected intravenously (via cephalic vein or femoral vein) using a peristaltic pump. 3 weeks later, animals were subject to transcardial perfusion with PBS, followed by 4% PFA. Tissues were then collected and processed for paraffin embedding and sectioning.

(91) 8. Immunohistochemistry

(92) Floating staining was performed for mouse tissue sections with primary antibodies diluted in PBS containing 10% donkey serum and 2% Triton X-100. Primary antibodies used include: chicken anti-GFP (1:1000); rabbit anti-RFP (1:1000); mouse anti-NeuN (1:500); rat anti-GFAP(1:500); Goat anti-GFAP(1:500); mouse anti-CD31(1:500). Secondary antibodies conjugated to fluorophores of Alexa Fluor 488, Alexa Fluor 555 or Alexa Fluor 647 were applied against the primary antibody's host species at a dilution of 1:200.

(93) For paraffin sections of NHP tissue, DAB staining was performed to visualize cells transduced by AAV-AADC. Rabbit anti-AADC antibody (1:500, Millipore) was used as primary antibody.

(94) 9. AAV Binding Assay

(95) HEK293T cells were cultured at 37° C. in a 5% CO₂ incubator. One day after seeding of HEK293T cells in a 24-well plate at a density of 250,000 cells per well, a cDNA plasmid of LY6A was transiently transfected into the cells using a transfection mixture of 200 µl DMEM (31053028; Gibco), 1 µg DNA plasmid and 3 µg of PEI. 48 hours post transfection, cells were placed on ice to chill down for 10 mins. The medium was then changed with 500 µl ice-cold serum-free DMEM medium containing rAAVs-mCherry at MOI of 10000. After incubating on ice for one hour, cells with presumably AAVs bound to their surface were washed with cold PBS for three times and were then subject to genomic DNA isolation. Cell-binding viral particles were quantified by using qPCR with primers specific to mCherry and normalized to HEK293T genomes using human GCG as reference.

(96) 10. Mouse Model of Glioblastoma

(97) All experiments were performed in compliance with protocols approved by the Animal Care and Use Committees (IACUC) at the Brigham and Women's Hospital and Harvard Medical School. Syngeneic immuno-competent C57BL/6 female mice weighing 20+/-1 g (Envigo) were used. GL261-Luc (100,000 mouse glioblastoma cells) resuspended in 2 µL phosphate buffered saline (PBS) was injected intracranially using 10 µl syringe with a 26-gauge needle (80075; Hamilton). A stereotactic frame was used to locate the implantation site (coordinates from bregma in mm: 2 right, 0.5 forward, at a depth of 3.5 into cortex). 7 days later, 200 µl AAV-HSV-TK1 (1E+12 viral genomes, IV) was administered once and ganciclovir (50 mg/kg) was administered daily for 10 days.

Example 1. Modification of AAV9 Capsid

(98) To identify peptide sequences that would enhance permeation of a biomolecule or virus across the blood brain barrier an AAV peptide display technique was used. Individual cell-penetrating peptides, as listed in Table 3, were inserted into the AAV9 capsid between amino acids 588 and 589 (VP1 numbering) as illustrated in FIG. 1A. The insertion was carried out by modifying the RC plasmid, one of the three plasmids co-transfected for AAV packaging; FIG. 1B shows an exemplary schematic of the experiments. Individual AAV variants were produced and screened separately. See Materials and Methods #1-3 for more details.

(99) TABLE-US-00006

TABLE	No.	Name	of CPP	Viral AAV	CPP	insert	Amino acid sequence	of CPP	# residues	titer	Initial AAV9	N/A	N/A	N/A	Normal screening	AAV.CPP.1	SynB1
RGGRLSYSRRRFSTSTGR	93	18	Low	AAV.CPP.2	L-2	HARIKPTFRRLKWKY	94	20	Low								
KGKFW	AAV.CPP.3	PreS2-TLM	PLSSIFSRIGDP	95	12	Low	AAV.CPP.4	Transportan									
AGYLLGKINKALAA	96	21	Low	10	LAKKIL	AAV.CPP.5	SAP	VRLPPP	VRLPPP	VRLPPP	97	18	Normal	AAV.CPP.6	SAP(E)	VELPPP	VELPPP
VELPPP	98	18	Normal	AAV.CPP.7	SVM3												
KGTYKKKLMRIPLKGT	99	16	Low	AAV.CPP.8	(PPR)3	PPRPPRPPR	100	9	Normal	AAV.CPP.9							

(PPR)5 PPRPPRPPRPPR 101 15 Low AAV.CPP.10 Polyarginine RRRRRRRR 102 8 Low AAV.CPP.11 Bip1 VPALR 1 5 Normal AAV.CPP.12 Bip2 VSALK 2 5 Normal AAV.CPP.13 DPV15 LRRERQSR LRRERQSR 103 16 NA AAV.CPP.14 HIV-1 Tat RKKRRQRRR 104 9 NA Follow-up AAV.CPP.15 Bip1.1 TVPALR (Rat) 3 6 Normal screening AAV.CPP.16 Bip2.1 TVSALK (Syn) 4 6 Normal AAV.CPP.17 Bip2.2 FTVSALK (Syn) 5 7 Normal AAV.CPP.18 Bip2.3 LTVSALK (Syn) 6 7 Normal AAV.CPP.19 Bip2.4 KFTVSALK (Syn) 72 8 Normal AAV.CPP.20 Bip2.5 TFVSALK (Syn) 7 7 Normal AAV.CPP.21 Bip2.6 TVSALFK (Syn) 8 7 Normal AAV.CPP.22 Bip2.6Rat TVPALFR (Rat) 9 7 Normal

Example 2. First Round of In Vivo Screening

(100) AAVs expressing nuclear RFP (H2B-RFP) were injected intravenously in adult mice with mixed C57B3L/6 and BALB/c genetic background. 3 weeks later, brain tissues were harvested and sectioned to reveal RFP-labelled cells (white dots in FIGS. 2A and 2C, quantified in FIGS. 2B and 2D, respectively). CPPs BIP1 and BTP2 were inserted into the capsids of AAV.CPP.11 and AAV.CPP.12, respectively. See Materials and Methods #4-5 for more details.

Example 3. Optimization of Modified AAV9 Capsids

(101) AAV.CPP.11 and AAV.CPP.12 were further engineered by optimizing the BIP targeting sequences. BIP inserts were derived from the protein Ku70 (See FIG. 3A and Material/Methods #1 for full sequence). The BIP sequence VSALK, which is of “synthetic” origin, was chosen as a study focus to minimize potential species specificity of engineered AAV vectors. AAVs were produced and tested separately for brain transduction efficiency as compared with AAV9 (see FIGS. 3B-C). Percentages of cell transduction in the mouse liver 3 weeks after IV injection of some AAV variants delivering the reporter gene RFP are shown in FIG. 3D. See Materials and Methods #1-5 for more details.

Example 4. In Vitro Model—BBB Permeation Screening

(102) Some of the AAV variants were screened for the ability to cross the human BBB using an in vitro spheroid BBB model. The spheroid contains human microvascular endothelial cells, which form a barrier at the surface, and human pericytes and astrocytes. AAVs carrying nuclear RFP as reporter were assessed for their ability to penetrate from the surrounding medium into the inside of the spheroid and to transduce the cells inside. FIG. 4A shows an experimental schematic. FIGS. 4B-D show results for wt AAV9, AAV.CPP.16, and AAV.CPP.21, respectively, those and other peptides are quantified in FIG. 4E. In this model, peptides 11, 15, 16, and 21 produced the greatest permeation into the spheroids. See Materials and Methods #6 for more details.

Example 5. In Vivo BBB Permeation Screening

(103) AAV.CPP.16 and AAV.CPP.21 were selected for further evaluation in an in vivo model, in experiments performed as described above for Example 2. All AAVs carried nuclear RFP as reporter. Both showed enhanced ability vs. AAV9 to transduce brain cells after intravenous administration in C57BL/6J adult mice (white dots in brain sections in FIG. 5A, quantified in FIG. 5B) and in BALB/c adult mice (white dots in brain sections in FIG. 6A, quantified in FIG. 6B).

(104) High doses of AAV.CPP.16 and AAV.CPP.21 (4×10^{12} vg per mouse, administered IV) resulted in widespread brain transduction in mice. Both AAVs carried nuclear RFP as reporter (white dots in brain sections in FIG. 7A, quantified in FIG. 7B).

Example 6. In Vivo Distribution of Modified AAVs

(105) As shown in FIG. 8A, AAV.CPP.16 and AAV.CPP.21 preferentially targeted neurons (labeled by a NeuN antibody) across multiple brain regions in mice including the cortex, midbrain and hippocampus. Both AAVs carried nuclear RFP as a reporter.

(106) AAV.CPP.16 and AAV.CPP.21 also showed enhanced ability vs. AAV9 in targeting the spinal cord and motor neurons in mice. All AAVs carry nuclear RFP as reporter and were administered intravenously into neonate mice (4×10^{10} vg). Motor neurons were visualized using CHAT antibody staining. Co-localization of RFP and CHAT signals in FIG. 8B suggested specific transduction of the motor neurons.

(107) The relative abilities of AAV-CAG-H2B-RFP and AAV.CPP.16-CAG-H2B-RFP to transduce

various tissues in mice was also evaluated. 1×10^8 vg was injected intravenously. The number of cells transduced was normalized to the number of total cells labeled by DAPI nuclear staining. The results showed that AAV.CPP.16 was more efficient than AAV9 in targeting heart (FIG. 9A); skeletal muscle (FIG. 9B), and dorsal root ganglion (FIG. 9C) tissue in mice.

Example 7. BBB Permeation in a Non-Human Primate Model

(108) 2×10^8 vg/kg AAVs-CAG-AADC (as reporter gene) were injected intravenously into 3-month-old cynomolgus monkeys. AAV-transduced cells (shown in black) were visualized using antibody staining against AADC. As shown in FIGS. 10A-D, AAV.CPP.16 and AAV.CPP.21 showed enhanced ability vs. AAV9 to transduce brain cells after intravenous administration in non-human primates. AAV.CPP.16 transduced significantly more cells than wt AAV9 in the primary visual cortex (FIG. 10A), parietal cortex (FIG. 10B), thalamus (FIG. 10C), and cerebellum (FIG. 10D). See Materials and Methods #7-8 for more details.

Example 8. AAV.CPP.16 and AAV.CPP.21 do not Bind to LY6A

(109) LY6A serves as a receptor for AAV.PHP.eB and mediates AAV.PHP.eB's robust effect in crossing the BBB in certain mouse strains. Over-expressing mouse LY6A in cultured 293 cells significantly increased binding of AAV.PHP.eB to the cell surface (see FIG. 11A). On the contrary, over-expressing LY6A does not increase viral binding for AAV9, AAV.CPP.16 or AAV.CPP.21 (see FIG. 11B). This suggests AAV.CPP.16 or AAV.CPP.21 does not share LY6A with AAV.PHP.eB as a receptor. See Materials and Methods #9 for more details.

Example 9. Delivering Therapeutic Proteins to the Brain Using AAV.CPP.21

(110) AAV.CPP.21 was used to systemically deliver the “suicide gene” HSV.TK1 in a mouse model of brain tumor. HSV.TK1 turns the otherwise “dormant” ganciclovir into a tumor-killing drug. Intravenously administered AAV.CPP.21-H2BmCherry (FIG. 12A, bottom left and middle right panel) was shown to target tumor mass, especially the tumor expanding frontier. As shown in FIGS. 12B-C, using AAV.CPP.21 to systemically deliver the “suicide gene” HSV.TK1 resulted in shrinkage of brain tumor mass, when combined with the pro-drug ganciclovir. These results show that AAV.CPP.21 can be used to systemically deliver a therapeutic gene into brain tumor. See Materials and Methods #10 for more details.

Example 10. Intracerebral Administration of AAV.CPP.21

(111) In addition to systemic administration (such as in Example 2), an AAV as described herein was administered locally into the mouse brain. Intracerebral injection of AAV9-H2B-RFP and AAV.CPP.21-H2B-RFP (FIG. 13) resulted in more widespread and higher-intensity RFP signal in AAV.CPP.21-treated brain sections vs. AAV9-treated ones. See Materials and Methods #4 for more details.

Other Embodiments

(112) It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

Claims

1. An AAV capsid protein comprising a targeting sequence that comprises TVSALFK (SEQ ID NO:8); TVSALK (SEQ ID NO:4); KLASVT (SEQ ID NO:83); or KFLASVT (SEQ ID NO:84), wherein the targeting sequence is inserted in a position corresponding to amino acids 588 and 589 of SEQ ID NO:85.
2. The AAV capsid protein of claim 1, comprising an amino acid sequence that comprises TVSALK (SEQ ID NO:4).
3. The AAV capsid protein of claim 1, comprising an amino acid sequence that comprises TVSALFK (SEQ ID NO:8).
4. The AAV capsid protein of claim 1, wherein the AAV is AAV9.

5. The AAV capsid protein of claim 1, comprising AAV9 VP1.
 6. An AAV comprising the capsid protein of claim 1.
 7. The AAV of claim 6, further comprising a transgene.
 8. The AAV capsid protein of claim 1, comprising an amino acid sequence of SEQ ID NO: 89 or SEQ ID NO: 90.
 9. The AAV of claim 7, wherein the transgene is a therapeutic transgene.
 10. The AAV of claim 7, wherein the transgene comprises a non-coding RNA.
 11. The AAV of claim 7, wherein the transgene encodes an anti-cancer agent.
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